Basic Math:

Digit:

Extraction of Digits

I/p N = 8221 ⇒

O/p 8, 2, 2, 1

N % 10 = 1

then

N/10 = 822

```
while (N>0){
    Last digit = N%10
    N = N/10
    Print
}
```

Count of digits: INT $\boxed{\log_{10}(N) + 1}$

Reverse a number:

```
while (N > 0) {
    Last digit = N % 10
        N = N/10
    reverse num = (reverse num x 10) + Last digit
```

Palindrome:

$$\boxed{\text{reverse num} == N}$$

Amstrong Num :-

$N = 371 \Rightarrow 3^3 + 7^3 + 1^3 = 371$
I/P

If N == cubed sum then true

Print all divisors:

```
for (i=1; i<=N; i++){
    if (N%i ==0){
        Print(i)
    }
}
```

TC: $O(N)$

(or)

```
for (i=1; i<=sqrt(N); i++){
    if (N%i ==0){
        Print (i)
        if (N/i != i){
            Print (N/i)
    }}
}
```

TC $O(\sqrt{n})$

If we need in order then use $\boxed{\text{List}}$

the $\boxed{TC\ O(\sqrt{n}) + O(n \log n)}$

Prime Numbers :-

↳ Exactly 2 factors 1 & its self.

count all divisors ≤ 2 true

> 2 false

1st method of no of divisors $TC = O(N)$

2nd method of no of divisors ⇒ $TC = O(\text{Sqrt}(N))$

GCD / HCF :-

$N_1 = 9$

1, 3, 9

$N_2 = 12$

1, 2, 6, 12, 3, 4

$GCD(9, 12) = 3$

then for 0 to min(N_1, N_2) {

if (N_1 % i == 0 && N_2 % i == 0)

gcd = i

TC = O(min(N_1, N_2)) = O(N)

(or)

for min(N_1, N_2) to 0 then

if (N_1 % i == 0 && N_2 % i == 0)

Print (i)

break

TC = O(N)

(or)

Euliden Algorithm for GCD:

gcd (N_1, N_2) = gcd ($N_1 - N_2$, N_2)

for $N_1 > N_2$

repeat

$$\gcd(20, 15) = \gcd(5, 15)$$
$$= \gcd(10, 5)$$
$$= \gcd(5, 5)$$
$$= \gcd(0, 5)$$

$\uparrow$

gcd.

(or)

$$\gcd(N_1, N_2) = \gcd(N_1 \% N_2, N_2)$$

$$TC = O(\log_{\phi} \min(N_1, N_2)) \quad N_1 > N_2$$

Imp points / notes :

1. Is prime :

```
if n == 1
    return false

for i = 2 ; i <= sqrt(n) ; i++
    if n % i == 0
        return false

return true
```

2. Is palindrome :-

If reverse num is getting more than max length of int then

rev num < = min length/10
rev num >= max length/10
return 0
rev num = (rev num x 10) + Last number